web_amrV2 deployment cheat sheet

Last updated: 2026-11-13 (Asia/Manila). This is an operational handoff, not a substitute for `deploy/README.md`. Inspect the target's actual state before changing anything; paths, versions and revisions below are observations, not guarantees.

Identify the application and targets

- Repository: `https://github.com/jundychitan/web_amrV2.git`; local Mac checkout: `/Users/rli/CodeIgniterProjects/web_amrV2`. This is the CodeIgniter project, not `camr_robinsons` (Laravel).
- Test: `172.104.42.220`, SSH user `root`, web app `http://172.104.42.220/amr/`. As last observed, the active release was `/srv/web-amr/releases/20261105-e854afe` through `/srv/web-amr/current`.
- MARIGOLD / production-side: `172.16.47.155`, hostname `mamasitas`, SSH user `amr`, web app `http://172.16.47.155/amr/`. Administrative operations previously used `su - root`, with a separate root password. Its observed active release was `/srv/web-amr/releases/test-reference-20261109` through `/srv/web-amr/current`.
- Both use Apache, MariaDB and PHP 8.4 FPM. Test last received gateway bulk upload commit `b54b8b7` on 2026-11-13. MARIGOLD last received RTU update support on 2026-11-12; the later meter/gateway bulk uploads were deployed only to test. Verify rather than assuming a whole release exactly matches a Git commit: changes were applied in place, and server-specific files were preserved.
- Credentials are deliberately omitted. Obtain current credentials through an approved channel; never print, commit, or place passwords in scripts. Verify SSH host keys through a trusted channel and keep host-key checking enabled.

VPN and connection

MARIGOLD requires the Cisco Secure Client VPN. The VPN server is `mama-sitas-twbnkdbzpgc.dynamic-m.com:443`. This hostname belongs in Cisco's **VPN connection** field, not the ThousandEyes Endpoint Agent connection-string screen. Ask the operator to complete sign-in/MFA as required.

On the Mac, check `/opt/cisco/secureclient/bin/vpn state`. A sandboxed check can falsely report that the service is unavailable; run a host-level check if needed. Confirm SSH connectivity separately:

```
ssh -o StrictHostKeyChecking=yes -o ConnectTimeout=8 amr@172.16.47.155
```

```
ssh -o StrictHostKeyChecking=yes -o ConnectTimeout=8 root@172.104.42.220
```

If the VPN is disconnected, ask the user to connect; do not infer credentials or disable security controls. A connection from the Mac does not establish network reachability from an actual gateway.

Runtime layout and constraints

- `/srv/web-amr/current` is the active application symlink, **not necessarily a Git checkout**. Do not `git pull` there or point it at a fresh clone. If a separate source checkout is needed, use a separate private directory, inspect it first, and keep deployment configuration and writable data intact.
- Web server config: `/etc/apache2/sites-available/001-web-amr.conf`; FPM pool: `/etc/php/8.4/fpm/pool.d/web-amr.conf`; socket `/run/php/web-amr.sock`. Protected FPM environment: `/etc/web-amr/web-amr.env`. On MARIGOLD, protected database client options include `/secure/web-amr/web_amr_app.cnf` and `web_amr_stage.cnf`. Never display their contents.
- Root owns deployed code; Apache/FPM run as `www-data`. Preserve existing owner, mode and symlinks. Writable runtime locations include `reports`, `meter_upload_list`, `application/logs`, `application/cache`, and `/var/lib/web-amr/sessions`. Do not replace server-specific database configuration, environment files, uploads, reports, logs, sessions, or data.
- The current deployments use HTTP by explicit request. Gateway endpoints are unauthenticated; restrict network access to approved sources. Credentials and readings are not encrypted over HTTP.

Update procedure

1. Confirm the user's requested target (test versus MARIGOLD) and scope. Check the working tree, fetch and inspect incoming commits. Fast-forward the local branch only if clean and requested; never overwrite uncommitted work.

   ```
   git status --short --branch
   git fetch origin main
   git log --oneline HEAD..origin/main
   git diff --stat HEAD..origin/main
   git pull --ff-only origin main  # only if appropriate
   ```

2. Review the delta and dependencies. Run appropriate tests; the full local suite includes a disposable localhost HTTP server, so the sandbox may need permission for loopback networking.

```
python3 tests/run_php84_tests.py --php "$(command -v php)"
node tests/meter_upload_ui_test.js
python3 tests/rtu_gateway_test.py
```

3. Inspect target state: `readlink -f /srv/web-amr/current`, service health, relevant file hashes and routes, PHP version/extensions, and any local changes. Compare each file to the known previous revision before overwriting. The test site's PHP FPM pool was configured with 90M `upload_max_filesize` and 100M `post_max_size`; workbook application limit is 5 MiB. Recheck actual values and extensions (`zip`, XML, `mbstring`) on the target.

4. Back up all affected existing files in a root-only location, recording files that were previously absent. Transfer only reviewed runtime changes; do not copy a full tree over a live release or replace protected configuration. Preserve existing server routes when merging new routes. Check syntax with `php8.4 -l`, `apache2ctl configtest` if Apache changed, and check file ownership/readability. A graceful `systemctl reload php8.4-fpm` makes PHP changes visible; reload Apache only after config validation. Observe output and service state. Keep a rollback plan for every changed file, including removal of newly added files.

5. Verify actual HTTP behavior and application workflows. Do not insert real meter or gateway data merely for smoke tests. When appropriate, use isolated disposable test records, a randomly named short-lived test account and synthetic workbooks; use preview/dry-run, delete test records in a guaranteed cleanup path, and verify deletion. Do not clear or commandeer an active user's session without explicit authorization. Never test flag reset against real gateways.

6. Report target, deployed commit/scope, tests, limitations, and backup path. Do not claim a complete Git deployment if only a selected delta was copied. Commit/push only when requested.

URLs and operational checks

- App: `http://<host>/amr/`.
- Meter readings: `http://<host>/http_post_server.php` (existing gateway request format); avoid posting fabricated readings to a live database.
- Time check: `http://<host>/check_time.php` on MARIGOLD. The test server previously lacked this route; inspect actual config.
- Gateway update script expects `http://<host>/rtu/index.php/rtu/rtu_check_update/<MAC>/<action>`, where action is one of `get_update_csv`, `get_content_csv`, `reset_update_csv`, `get_update_location`, `get_content_location`, `reset_update_location`. This needs both an Apache root rewrite and a CodeIgniter route in `application/config/routes.php`. Check status with a known

nonexistent MAC only after confirming it is not registered. The legacy shell script uses curl without `--fail`; an HTTP error while downloading may overwrite a gateway config file. Test with a real gateway only when the operator is ready.
- Meter and gateway upload pages are under Maintenance. The APIs are `/amr/index.php/api/meters/import[/token]` and `/amr/index.php/api/gateways/import[/token]`; unauthenticated token requests should return 401. For actual behavior use an authorized session and preview (dry-run), not a production import.

```
systemctl is-active apache2 php8.4-fpm mariadb
apache2ctl configtest
curl -i http://<host>/amr/index.php/api/gateways/import/token
```

Rollback and session locks

- Test backup before gateway bulk upload: `/root/amr-gateway-backup-20261113175351`; before meter bulk upload: `/root/amr-meter-backup-20261113160130`.
- MARIGOLD backup before RTU rollout: `/secure/web-amr/rollback-rtu-20261112143215`. These backups contain only files affected by their respective deployment, not database snapshots or full release images. Inspect contents first, restore the appropriate saved paths and remove files that did not exist previously, validate config and reload services. Avoid rolling back later unrelated changes.
- Admin accounts have a database-backed single-session lock. Clearing `user_tb.active_session_hash` and `active_session_activity` logs out the account's existing session; do this only when authorized, for the exact account, and check that it is unique. If it immediately returns, another login may be active. Do not print password hashes or database secrets. MARIGOLD also has an `eng` administrator account created at the user's request; do not assume its credentials or session state remain unchanged.